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CLASS/SEC:5-E
SRN:PES1UG24AM251

USE CASE NO:24

Req ID
Type Description
Priorit y
Acceptance Criteria
Rationale

FR-0 01

FR-0 02

FR-0 03

FR-0 04

FR-0 05
Functional The system shall collect bus GPS

coordinates every 5 seconds and update the estimated arrival times for upcoming stops.

Functional The system shall allow commuters to view the current crowding status of a bus.

Functional The system shall allow the Fleet Controller to monitor the current location and crowding status of buses.

Functional The system shall allow commuters to view the estimated arrival time of a bus at their selected stop.

Functional The system shall display the

computed passenger crowding level for each bus High ETA is recalculated using the latest GPS information

and changes appropriately as the bus moves.

High When a commuter selects a bus, its current crowding level is displayed.

High When the Fleet Controller selects a bus, its GPS location and crowding status are displayed.

High When the commuter selects a bus and stop, the system displays the estimated arrival time.

High When updated ticketing data is received, the system computes and updates the Helps commuters know when the bus will arrive.

whether to take the bus.

and reduce waiting time.

Helps commuters decide

Helps the controller monitor bus operations.

Helps commuters plan their journey

Keeps crowding information up to date.

ticketing sensor data.

seconds.

NFR -001

The tracking engine shall handle telemetry streams from up to 1,000 active transit vehicles simultaneously.

High Benchmark testing confirms that the system handles 1,000 active vehicles under the required performance conditions.

Ensures the system can support a large city-wide fleet.

NFR -002
Nonfuncti onal

The system shall load and display the latest bus tracking information within 5 seconds of a commuter's request.
bus's crowding status.

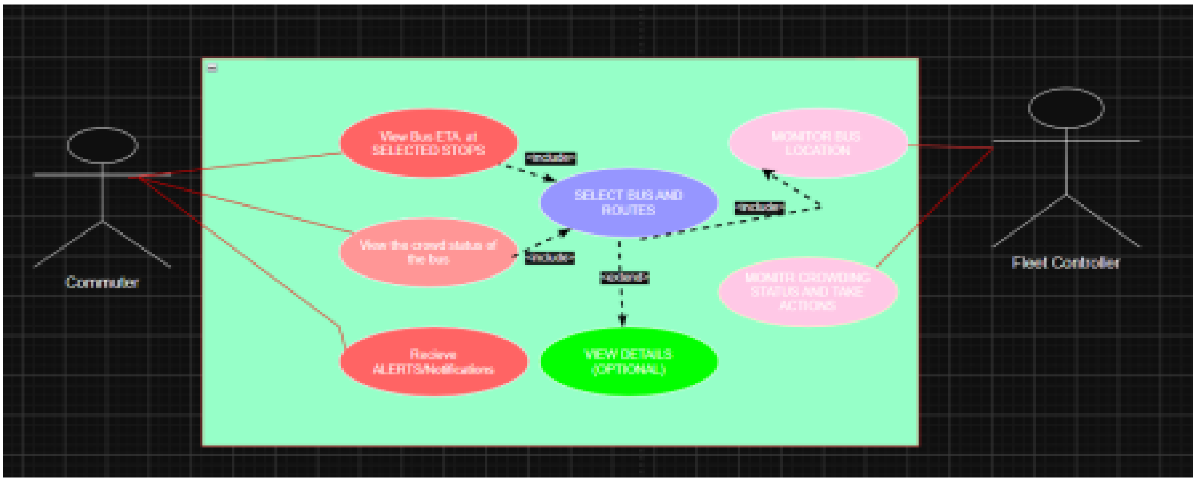
High When the

commuter requests the latest information, the system displays the updated location, ETA, and crowding information within 5

Ensures commuters receive timely live information.

Nonfuncti onal
based on

USE CASE DIAGRAM



Use-Case Flow – Track Bus and View ETA

Preconditions

- The bus routes should be available in the system.
- The buses should be sending their GPS location.

Main Success Scenario

1. The commuter searches for the bus route they want. 2. The commuter selects the bus.
3. The system shows where the bus is currently located. 4. The system shows how crowded the bus is.
5. The system shows the estimated time for the bus to reach the selected stop.
6. The commuter looks at this information and decides whether they want to take the bus.

Postcondition

The commuter gets the bus location, ETA, and crowding information and can decide whether to take the bus.

Alternate Flow – GPS Data Not Available

1. The commuter selects a bus route.
2. The system is not able to get the GPS information of the bus. 3. The system shows a message saying, “Sorry! Unable to fetch bus tracking data.”
4. The commuter can try again or select another bus.